

Day 27: Linear Algebra - Full Explanation Notebook

DETAILED SCENARIOS, COFACTOR BREAKDOWNS & FULL SOLUTIONS

1. Core Concepts: Vectors and Matrices

English: Vectors represent single-dimensional arrays used to store sequential feature states, whereas Matrices are multi-dimensional data tables structured with horizontal rows and vertical columns that simulate full database architectures.

Vectors matlab ek dimensional array hote hain jisme ek line ka sequence data hota hai, jabki Matrices multi-dimensional tables hoti hain jisme horizontal rows aur vertical columns poore database structural design ko darshate hain.

2. Architectural Importance in Machine Learning

English: Standard matrix operations bypass iterative loops by executing synchronized arithmetic transformations across multidimensional data layers simultaneously. This framework is essential for parameter optimization and neural network processing.

Matrix operations computer ke andar alag-alag loops chalane ke bajaye sabhi dimensions par ek sath parallel processing chalte hain. Yeh parameters optimization aur neural networks ke liye sabse zyada zaroori framework hai.

English: Historically developed across more than a century by pioneers like Gabriel Cramer (1750), Carl Friedrich Gauss (1800s), and Arthur Cayley (1850s), this system reduces complex calculations into clean mathematical structures designed for computational execution.

Itihaas ke mutabik 100 se 150 saal ke dauran Gabriel Cramer (1750), Carl Friedrich Gauss (1800s), aur Arthur Cayley (1850s) ne is pure system ko develop kiya taaki bade computations ko computer ke anukul banaya ja sake.

3. Comprehensive Inversion Problems with Scenarios

Example 1: The 2x2 Showroom Performance Matrix

Scenario / Question: At Goda Gold, a branch tracks two customer metrics: Silver buyers (x) and Gold buyers (y). Equation 1 represents item count: 3 items per silver buyer and 2 items per gold buyer equals 12 items. Equation 2 tracks total processing points: 4 points per silver buyer and 5 points per gold buyer equals 23 points. Find the system model parameters by finding the exact Inverse Matrix of the system rules.

Goda Gold showroom par do tarah ke buyers hain: Silver buyers (x) aur Gold buyers (y). Pehli equation items ki hai: har silver buyer 3 items aur gold buyer 2 items leta hai, kul 12 items bike. Doosri equation points ki hai:

har silver buyer ko 4 aur gold buyer ko 5 points milte hain, kul 23 points mile. System rules ka exact Inverse Matrix nikal kar model parameters find karein.

System Configuration Matrix A:

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$$

Step 1.1: Calculate Determinant (|A|)

Cross multiply the main diagonal and subtract the off-diagonal components:

Main diagonal ko cross multiply karein aur baaki elements ko minus karein:

$$|A| = (3 \times 5) - (2 \times 4) = 15 - 8 = 7$$

Step 1.2: Find Co-factors and Adjugate Matrix (adj A)

For a 2x2 matrix, exchange positions of A11 and A22, and multiply A12 and A21 by -1:

2x2 matrix ke liye A11 aur A22 ki jagah badlein, aur A12 aur A21 ko -1 se multiply karein:

$$C_{11} = +5, C_{12} = -4, C_{21} = -2, C_{22} = +3$$

$$\text{Cofactor Matrix } C = \begin{bmatrix} 5 & -4 \\ -2 & 3 \end{bmatrix}$$

Transpose of C gives Adjugate Matrix adj(A):

Cofactor Matrix ka Transpose lene par Adjugate Matrix milta hai:

$$\text{adj}(A) = \begin{bmatrix} 5 & -2 \\ -4 & 3 \end{bmatrix}$$

Step 1.3: Apply Inverse Formula ($A^{-1} = \text{adj } A / |A|$)

Divide every internal scalar element by the determinant value (7):

Adjugate ke har element ko determinant value (7) se divide karein:

$$A^{-1} = \begin{bmatrix} 5/7 & -2/7 \\ -4/7 & 3/7 \end{bmatrix} = \begin{bmatrix} 0.7143 & -0.2857 \\ -0.5714 & 0.4286 \end{bmatrix}$$

Example 2: The 3x3 Daily Multi-Counter Transaction Matrix

Scenario / Question: A regional retail store sets three separate counter audits monitoring Ring sales (x), Chains (y), and Coins (z). The operational flow coefficient values build Matrix A. Complete a granular calculation of the Inverse Matrix to establish an automated accounting engine.

Ek regional retail store teen alag-alag counters par Rings (x), Chains (y), aur Coins (z) ki sales check karta hai. Iska operational flow coefficient Matrix A banata hai. Iska poora full step Inverse Matrix nikalie taaki accounting engine set kiya ja sake.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Step 2.1: Granular Expansion of Determinant ($|A|$) through Row 1

$$|A| = 1 \times \det[[1, 4], [6, 0]] - 2 \times \det[[0, 4], [5, 0]] + 3 \times \det[[0, 1], [5, 6]]$$

$$\det1 = (1 \times 0 - 4 \times 6) = -24 \mid \det2 = (0 \times 0 - 4 \times 5) = -20 \mid \det3 = (0 \times 6 - 1 \times 5) = -5$$

$$|A| = 1(-24) - 2(-20) + 3(-5) = -24 + 40 - 15 = 1$$

Row 1 ke along minor determinants expand karne par final Determinant 1 aaya.

Step 2.2: Full Step Calculation of All 9 Individual Cofactors

- $C_{11} = +((1 \times 0) - (4 \times 6)) = -24$
- $C_{12} = -((0 \times 0) - (4 \times 5)) = -(-20) = 20$
- $C_{13} = +((0 \times 6) - (1 \times 5)) = -5$
- $C_{21} = -((2 \times 0) - (3 \times 6)) = -(-18) = 18$
- $C_{22} = +((1 \times 0) - (3 \times 5)) = -15$
- $C_{23} = -((1 \times 6) - (2 \times 5)) = -(6 - 10) = 4$
- $C_{31} = +((2 \times 4) - (3 \times 1)) = 8 - 3 = 5$
- $C_{32} = -((1 \times 4) - (3 \times 0)) = -4$
- $C_{33} = +((1 \times 1) - (2 \times 0)) = 1$

Assembled Cofactor Matrix C:

$$C = \begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix}$$

Step 2.3: Transpose Execution to get Adjugate Matrix (adj A)

Cofactor Matrix ke rows ko columns mein convert (transpose) karne par:

$$\text{adj}(A) = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Step 2.4: Final Division by Determinant ($A^{-1} = \text{adj } A / 1$)

$$A^{-1} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Example 3: The 4x4 Enterprise Level Consumer Segment Matrix

Scenario / Question: Goda Gold Dubai showroom processes four dynamic large-scale streams: Retail (x), Corporate (y), VIP Investors (z), and Tourists (w). The metrics matrix is assembled below based on total items, making charges revenue, reward points, and interaction time. Perform a full multi-stage calculation to determine the exact matrix inverse parameters.

Goda Gold Dubai showroom char alag customer segments handle karta hai: Retail (x), Corporate (y), VIP (z), aur Tourists (w). Items count, making charges, rewards points aur time ke records se ye 4x4 Matrix bana hai. Iska full step multiplication aur inversion process nikalie.

$$A = \begin{bmatrix} 2 & 10 & 25 & 5 \\ 100 & 500 & 2000 & 300 \\ 1 & 4 & 10 & 2 \\ 15 & 60 & 120 & 30 \end{bmatrix}$$

Step 3.1: Complete 4x4 Determinant Expansion Loop

$$|A| = 2 \cdot \det(M11) - 10 \cdot \det(M12) + 25 \cdot \det(M13) - 5 \cdot \det(M14)$$

Let's evaluate each sub 3x3 determinant systematically:

- **det(M11):** Row 1, Col 1 removed. Yields $\begin{bmatrix} 500 & 2000 & 300 \\ 4 & 10 & 2 \\ 60 & 120 & 30 \end{bmatrix}$. Value = -6000
- **det(M12):** Row 1, Col 2 removed. Yields $\begin{bmatrix} 100 & 2000 & 300 \\ 1 & 10 & 2 \\ 15 & 120 & 30 \end{bmatrix}$. Value = -3000
- **det(M13):** Row 1, Col 3 removed. Proportional linear rows found inside. Value = 0
- **det(M14):** Row 1, Col 4 removed. Yields $\begin{bmatrix} 100 & 500 & 2000 \\ 1 & 4 & 10 \\ 15 & 60 & 120 \end{bmatrix}$. Value = 3000

$$|A| = 2(-6000) - 10(-3000) + 25(0) - 5(3000) = -12000 + 30000 - 15000 = 3000$$

Pure calculation ke baad system ka Determinant value 3000 aaya.

Step 3.2: Full Adjugate Configuration (adj A) via Transposed Cofactors

Evaluating all 16 multi-dimensional matrix dimensions gives the transposed configuration matrix:

Sabhi 16 coordinates ke minors aur signs analyze karne ke baad Adjugate Matrix milta hai:

$$\text{adj}(A) = \begin{bmatrix} -6000 & 0 & 15000 & 0 \\ 3000 & -30 & 4500 & -500 \\ 0 & 0 & 1500 & -100 \\ -3000 & 60 & -22500 & 1500 \end{bmatrix}$$

Step 3.3: Final Structural Matrix Inverse Layout ($A^{-1} = \text{adj } A / 3000$)

Divide every standalone integer by 3000 to extract the clean parametric array:

Adjugate ke har ek part ko 3000 se divide karne par sateek model rules milte hain:

$$A^{-1} = \begin{bmatrix} -2 & 0 & 5 & 0 \\ 1 & -0.01 & 1.5 & -0.1667 \\ 0 & 0 & 0.5 & -0.0333 \\ -1 & 0.02 & -7.5 & 0.5 \end{bmatrix}$$